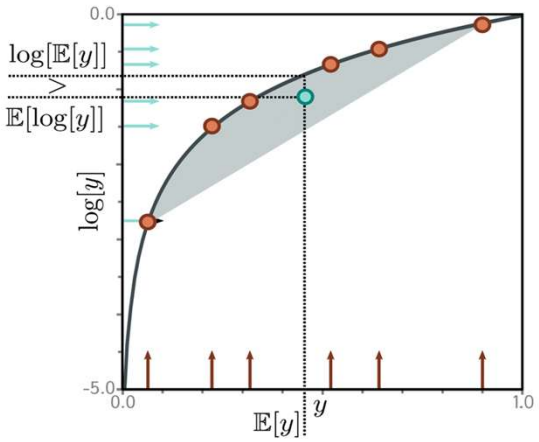
# Agenda

- Last time
  - Inequalities (Markov, Chebyshev, Jensen)
- Today

Illustration for log function




$$E\left[\log[y]\right] \leq \log\left[E[y]\right]$$

2

# Markov's Inequality Discrete Case

$\Pr[x>a]$

$\leq \frac{E[x]}{a}$

$$a\Pr(X > a)$$

$$= a\Pr(X = a+1) + a\Pr(X = a+2) + a\Pr(X = a+3) + \cdots$$

$$\leq a\Pr(X = a) + (a+1)\Pr(X = a+1) + (a+2)\Pr(X = a+2) + (a+3)\Pr(X = a+3) + \cdots$$

$$\leq \Pr(X = 1) + 2\Pr(X = 2) + 3\Pr(X = 3) + \cdots +$$
$$a\Pr(X = a) + (a+1)\Pr(X = a+1) + (a+2)\Pr(X = a+2) + (a+3)\Pr(X = a+3) + \cdots$$

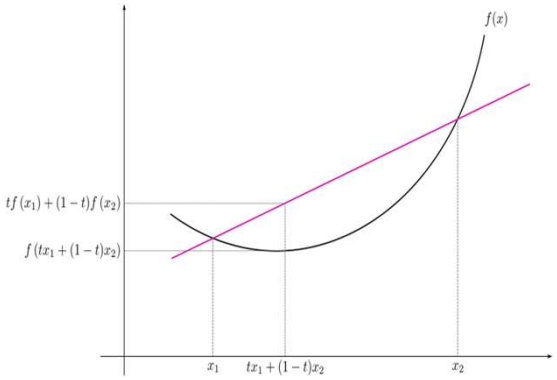
$$= E(X)$$

3

3

# Jensen's inequality

- For any concave function and random variable Y
  - $f[E[\mathrm{Y}]] \geq E[f[\mathrm{Y}]]$
  - For a convex function
    - it's the other way around




4

# Example



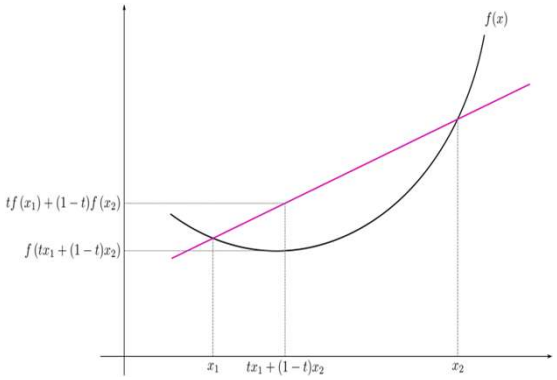
- $R = \begin{cases} 1.1 & \text{with probability } \frac{1}{2} \\ 0.9 & \text{with probability } \frac{1}{2} \end{cases}$ Interpret R as return on investment
  - E[R] = 1 suggests that there is no gain or loss in investment
- However, net return $\mathrm{N} = R_1 R_2 R_3$
  - To find expected value of wealth, without assumption of independence, use
    - $\log N = \log R_1 + \log R_2 + \cdots$
    - $E[\log N] = \mathrm{E}[\log R_1] + \mathrm{E}[\log R_2] + \ldots$
    - From Jensen's formula on each term, LHS < log E[$R_1$] + log E[$R_2$] + ..
      - If each return was, on an average 1, then LHS < n log E[R] = n log 1 = 0
      - Turning to the actual wealth $\exp\{E[\log N]\} < e^0 = 1$ return factor is negative
- Conclusion: It's a losing proposition, not a no-op proposition

Used in English sense
-ve ⇒ not a good situation.
mathematically < 1
not < 0

5

# Agenda

- Last time
  - Inequalities
- Today
  - Weak Law of “large numbers”
  - Popular distribution functions




6

# Expected value

- When we say the expected value of a die roll is 3.5, what does that mean?
  - If you throw the die n times and find the sample mean, you should get a value close to 3.5 — provided n is large.
  - It says nothing about small n.
  - As n increases, the sample mean should move closer and closer to 3.5.
- That is the intuition. Let’s state it more formally

7

7

# The weak law of large numbers

- Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables, each with mean $\mu$. For any $\varepsilon > 0$:

$$P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \to 0 \text{ as } n \to \infty$$

- The **sample mean** converges in probability to the true mean

Must know

8

8

# The weak law

$$P\{|X - \mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

- From Chebyshev's inequality applied to the sample mean

$$E\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \mu, \quad \text{Var}\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

$$\therefore\ P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \leq \frac{\sigma^2}{n\varepsilon^2}$$

- Letting n → ∞ the right-hand side vanishes, so

$$\therefore\ \lim_{n\to\infty} P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} = 0$$

9

9

# Comments on the weak law

Nice to know

- The identically distributed assumption is not strictly necessary
  - The random variables just need the same mean $\mu$.
  - We assume a common variance earlier
    - If not, the $n\sigma^2$ term becomes n times the expected variance
    - The proof still goes through.
- Independence is strictly not necessary
  - Pairwise independence suffice
- The law *does* assume a well-defined expected value and variance
  - Otherwise, it may not hold.

10

10

# Law of averages

- We tend to believe that if something has been going wrong for a while, it will suddenly turn right — by the "law of averages"
  - This supposed law is a fallacy
  - A gambler tosses an unbiased coin 20 times and gets a head every time
    - He invokes the "law of averages" and concludes the next toss is more likely to be a tail
    - The probability of getting 21 heads in a row is $(1/2)^{21}$
    - The probability of getting 20 heads followed by 1 tail is also $(1/2)^{21}$
  - The coin has no memory: Events are independent with identical distribution
    - However, if he keeps tossing the coin eventually he will see tails appearing, but the tails may not "compensate" for the first 20 heads
    - Relative frequency stabilize, past imbalance need not be cancelled

12

12